Test Execution and Code Coverage Report

1. Overview

This report summarizes the unit testing and branch coverage analysis executed for the **Bus Transit Fare** module (Lab 6). Both test execution and code coverage measurement completed with a 100% success and coverage rate.

2. Execution Environment & Configuration

- **Operating System / Platform:** Windows 32-bit (win32)
- **Python Version:** 3.14.3
- **Test Framework:** pytest 9.2.0 (with pluggy 1.7.0)
- **Installed Plugins:** anyio 4.14.0, Faker 40.40.0, platformdirs 4.13.0
- **Working Directory:** C:\Users\sanya\Downloads\L6_L7_10_bus_transit_fare-main\Lab 6
- **Execution Commands:**

  PowerShell

  python -m coverage run --branch -m pytest test_bus_transit.py
  python -m coverage report -m
  ```[cite: 1]

## 3. Test Suite Results

- **Target Test File:** test_bus_transit.py
- **Total Tests Collected:** 16
- **Passed:** 16 (100%)
- **Failed:** 0
- **Execution Duration:** 0.47 seconds

## 4. Code & Branch Coverage Summary

| File Name | Statements | Missed | Total Branches | Partial Branches | Coverage |
|---|---|---|---|---|---|
| bus_transit_fare.py | 57 | 0 | 40 | 0 | **100%** |
| test_bus_transit.py | 67 | 0 | 0 | 0 | **100%** |
| **Total** | **124** | **0** | **40** | **0** | **100%** |

- **Statement Coverage:** Full coverage achieved (all 124 statements executed across implementation and test suites).
  ```

- **Branch Coverage:** All 40 decision branches within bus_transit_fare.py were evaluated in both true and false directions with 0 partial branches missed.

5. Warnings & Observations

- **Deprecation Notice:** A single non-blocking warning was flagged during execution (PytestRemovedIn10Warning):
    - **Cause:** anyio\pytest_plugin.py references _pytest.python.CallSpec2, which has been renamed to CallSpec and is slated for removal in pytest 10.
    - **Action Item:** Updating the anyio package to a newer release will resolve this deprecation notice before pytest updates to major version 10.

Line coverage

```
PS C:\Users\sanya\Downloads\L6_L7_10_bus_transit_fare-main\Lab 6> python -m coverage run --branch -m pytest test_bus_transit
.py
>> python -m coverage report -m
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\sanya\Downloads\L6_L7_10_bus_transit_fare-main\Lab 6
plugins: anyio-4.14.0, Faker-40.40.0, platformdirs-4.13.0
collected 16 items

test_bus_transit.py ................                                                                 [100%]

============================================== warnings summary ===============================================
..\..\..\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14
  C:\Users\sanya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warni
ng: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================== 16 passed, 1 warning in 0.47s ==========================================
Name                 Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------
bus_transit_fare.py      57      0     40      0   100%
test_bus_transit.py      67      0      0      0   100%
----------------------------------------------------------------
TOTAL                   124      0     40      0   100%
PS C:\Users\sanya\Downloads\L6_L7_10_bus_transit_fare-main\Lab 6>
```

Branch coverage

```
PS C:\Users\sanya\Downloads\L6_L7_10_bus_transit_fare-main\Lab 6> python -m coverage run --branch -m pytest test_bus_transit
.py
>> python -m coverage report -m
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\sanya\Downloads\L6_L7_10_bus_transit_fare-main\Lab 6
plugins: anyio-4.14.0, Faker-40.40.0, platformdirs-4.13.0
collected 16 items

test_bus_transit.py ................                                                                 [100%]

============================================== warnings summary ===============================================
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  C:\Users\sanya\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warni
ng: _pytest.python.CallSpec2 has been renamed to CallSpec.
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    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
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test_bus_transit.py      67      0      0      0   100%
----------------------------------------------------------------
TOTAL                   124      0     40      0   100%
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```